Claude Code for Accounting Research

Module 2 — The Empirical Loop on SEC FSDS

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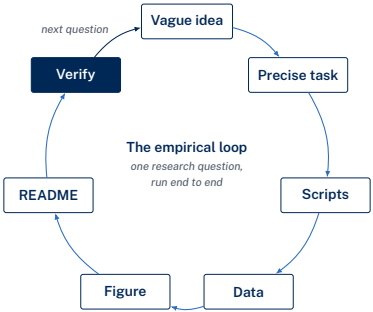
Yale School of Management

Today's Arc

- **0:00–0:40** Lecture: dataset anatomy, the loop, reproducibility habits
- **0:40–1:15** Live demo: R&D intensity by industry, empty folder to figure
- **1:15–1:25** Break
- **1:25–2:45** Lab: your own measure, empty folder to verified figure
- **2:45–3:05** Debrief, exit ticket, **WRDS checkpoint (hard gate)**

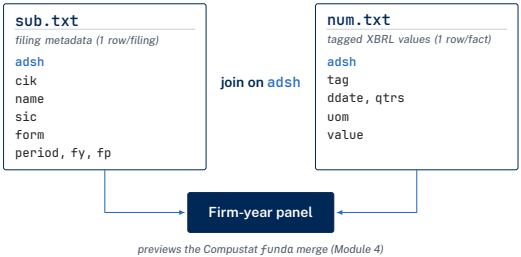
Recap: Module 1's Model, Now Tested

- Module 1: files persist, trust but verify, be specific — one script, one number
- Module 2: one accounting question, the *full* pipeline, end to end
- Every stage is a file on disk, not a chat reply



SEC FSDS: Two Files, One Key

- `sub.txt` — one row per filing: `adsh`, `cik`, `sic`, form, period
- `num.txt` — one row per tagged value: `adsh`, `tag`, `ddate`, `qtrs`, value
- Join key: `adsh` — state row counts before and after



Free and quarterly, with no credentials needed. Structured the way Compustat is.

Profile the Data Before You Plot It

- `DATA_PROFILE.md` — what one row of each table is, and what can go wrong with it
- Requested in the opening prompt, next to `DECISIONS.md` and `LOG.md`
- Written before any measure is computed; read by every later session
- Catches the errors a clean-looking figure hides

Applies "files persist, context doesn't" to the data layer. Module 4 asks the same questions of Compustat.

What the FSDS Profile Answers

- **What is one row?** `sub.txt`: one filing. `num.txt`: one value — tag, date, quarters, unit, filing
- **Is it unique?** Key columns from the SEC's FSDS documentation, then test: rows vs. distinct key combinations
- **Tag coverage?** Share of 10-Ks reporting each tag, per pinned quarter
- **Known problems?**
  - Prior-year comparatives: the same firm-year value appears in more than one filing
  - `10-K/A` amendments restate earlier values
  - For each, write down which filing you keep

If a third of filers never report a tag, a measure built on it reflects who reports it as much as the economics.

Vague Idea → Precise Task, Again

- "How has R&D intensity evolved across industries?" is a good research question but not a usable prompt
- Module 1's four components: file, operation, aggregation, output
- Today: a multi-file pipeline adds two more decisions, the join and the paper trail

Decomposing the R&D Question

- **Which files?** which FSDS quarters, which two files inside each
- **Which join?** on what key, checked how
- **Which measure?** R&D over assets — which XBRL tags, filtered how
- **Which aggregation?** what statistic, by what grouping, over what time unit
- **Which output?** what figure, saved under what name, alongside what table

Reproducibility as Code, Not Chat Lore

- **Empty-folder discipline** — `mkdir ~/lab2 && cd ~/lab2 && claude`
- **Named-script discipline** — delete the chat: does the pipeline still run?
- **Paper-trail prompting** — request `DATA_PROFILE.md` / `DECISIONS.md` / `LOG.md` / attrition rows *before* any code runs

The Opening-Prompt Template: Six Parts

DATA: *FSDS quarters,* `sub.txt` + `num.txt`*, columns named*

JOIN: *merge on* `adsh`*; keep consolidated rows (*`segments`*,* `coreg` *blank); row counts before/after; flag unmatched*

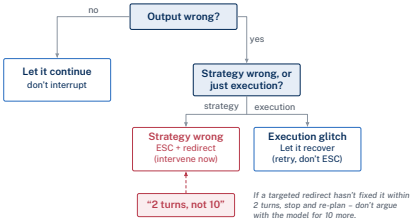
MEASURE: *R&D expense / total assets; tags named;* `qtrs` *handling; form filter*

AGGREGATION: *median by SIC2 $\times$ year, across all quarters*

OUTPUT: `rd_intensity.png` + `rd_intensity_by_sic2_year.csv`*, named in advance*

PAPER TRAIL: *named scripts only;* `DATA_PROFILE.md`*;* `DECISIONS.md`*;* `LOG.md`*; attrition row every sample-size change*

Iteration and Recovery



- The first figure is almost never right. That is normal.
- Small targeted rounds: axes and labels, then bucket granularity. Count them.
- Watch autonomous recovery (the 403); intervene only on strategy

Sanity Checks: Two, Minimum

- **Benchmark comparison** — pharma/tech should beat utilities/retail; does it?
- **Plausibility check** — does the series respond to the 2020 disruption?
- Name both checks *before* the figure renders
- Before either: plot the raw series (next slide)

Plot the Raw Series First

- Firm-level ratio, every observation, by fiscal year — before aggregating
- Once before winsorizing, once after
- Shows what a median hides:
  - R&D larger than total assets
  - A year where the observation count collapses
  - Exact zeros that should be missing
- Does not replace the two named checks. It tells you whether the series is worth summarizing

Common Failure

Treating a plausible-looking figure as a verified one

Clean axis labels and smooth lines say nothing about whether the tag selection, filter, or join was correct. A figure can look finished without being verified.

Verify This

Name your two checks before you look at the figure

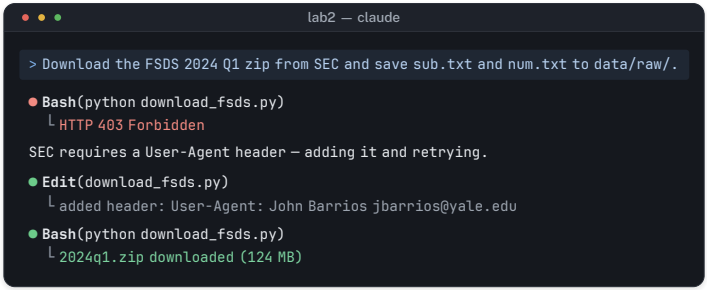
Decide the benchmark comparison and the plausibility check *before* the figure renders. Write both, with sources, into the README.

The Three Mantras, Carried Forward

Same three sentences, applied to more work

1. Files persist; context doesn't — DECISIONS.md, LOG.md, attrition rows, at every stage
2. Trust, but verify — two named sanity checks, not a vague sense it looks fine
3. Be specific — six parts now: data, join, measure, aggregation, output, paper trail

What You'll See: Error Recovery



Stylized session transcript (not a screenshot).

Autonomous error recovery: read the error, fix the cause, retry — no prompt from you.

The agent diagnoses and fixes a bare 403 Forbidden without you typing any of the fix. Watch rather than drive.

Next: Live Demo (0:40–1:15)

- Opening prompt read aloud, slowly: the template to keep (8 min)
- Download + 403 recovery: watch rather than drive (5 min)
- Clean + join on `adsh`; attrition rows print live (7 min)
- First-draft figure $\rightarrow$ two refinement rounds (8 min)
- Sanity checks, sign off, README written last (7 min)

What You'll See: Prompt and Join

- Six-part prompt projected and read aloud slowly, not retyped
- The SEC's User-Agent 403 and the agent's own recovery
- Row counts before/after the join; unmatched `adsh` flagged; one `DECISIONS.md` entry read aloud live

What You'll See: Iteration and Sign-Off

- The first draft renders wrong on purpose; we count the rounds together
- Round 1: axes and labels. Round 2: industry-bucket granularity
- Both sanity checks run and narrated; README written only after they pass

Next: The Lab (1:25–2:45)

- Empty folder $\rightarrow$ verified figure + README, one continuous 80-minute block
- Choose one measure: **base** leverage / **standard** R&D variant / **stretch** revenue concentration
- Different from the demo variable, so the verification is your own
- Requirements: named scripts, DECISIONS.md + LOG.md requested upfront, attrition table, one figure

Lab Verify Checklist

- ☐ Pipeline reruns start-to-finish from the README, in a fresh clone
- ☐ `sub`/`num` joined on `adsh` — row counts stated before/after
- ☐ Attrition table present; every sample-size change explained in `DECISIONS.md`
- ☐ `DATA_PROFILE.md`: row definition, uniqueness test, tag coverage, known problems
- ☐ Two sanity checks written down, with sources; raw series plotted
- ☐ Figure produced by a script — no interactive-only outputs

WRDS Checkpoint — Hard Gate

Due by the end of this module: submit one of the two

1. A `tunnel_status` screenshot: `ssh wrds` → Duo push → success
2. A flag on your exit ticket describing what went wrong

- Triaged tonight; fixes announced Module 3 morning
- Two days is the minimum lead time to fix an account or Duo problem
- Not working at this checkpoint → provisionally on the public-data fallback track for Module 4; get it working before Module 4 and you can switch to live

Debrief and Exit Ticket

- Whose first-draft figure was actually right? (Expect close to zero. That is normal.)
- What did Claude recover from alone; where did you have to step in?
- Exit ticket: standard 3 questions + sub-agent teaser (why might one have appeared today?)
- **WRDS checkpoint submitted — see previous slide**